DATA 1030 Midterm Report

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GitHub Repository: `https://github.com/khyue04/DATA1030-NoMissingValues`

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1 Introduction

Getting a mortgage is one of the most important financial steps for many people in the US, but the approval process can be time-consuming and uncertain. Lenders consider several factors when reviewing applications, including the applicant's income, existing debt, and the amount of the requested loan. The Home Mortgage Disclosure Act (HMDA) provides a valuable source of data for studying mortgage lending decisions. By examining these records, researchers can better understand which factors are associated with mortgage approval and denial.

This research goes back decades. In 1996, Munnell et al. [1] studied mortgage applications in Boston and found that applicants' finances, credit histories, and properties help explain denials, although minority applicants were still denied more often than White applicants with similar profiles. More recently, García and Garriga [2] analyzed more than 30 million home-purchase mortgage applications from 2018 to 2024. They found that debt-to-income ratio was the most commonly reported reason for denial and that denial rates rose sharply when this ratio exceeded 50%. Denial rates also increased in 2022 and 2023 as interest rates climbed. Machine learning has also been applied to mortgage lending decisions. Babaei et al. [3] used explainable machine learning models on New York State HMDA data to study lending decisions. They noted that rejected applications were much less common than approved ones, posing a challenge for model prediction and evaluation.

Our goal is to predict whether a mortgage application in Rhode Island will be denied, using HMDA data from 2018 to 2024. We will train and compare several machine learning models on loan and applicant information, then check which features contribute most to their predictions. Using data from different years also allows us to examine how applicant and loan characteristics have changed over time. We will further test whether models trained on earlier years can predict outcomes in later years.

2 Data

We use publicly available Home Mortgage Disclosure Act (HMDA) data from the Consumer Financial Protection Bureau HMDA platform. Covered financial institutions report information about mortgage applications and their outcomes, and the public data are modified to protect applicants'

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and borrowers' privacy. Our study focuses on home-purchase applications in the State of Rhode Island from 2018 to 2024.

We collected the data using Python's requests package and the HMDA Data Browser's API. We requested each year separately, specifying Rhode Island through states = 'RI' and home purchases through loan_purposes = '1'. We saved each successful response as a CSV file, loaded the files using pandas, checked their reported years and required columns, and combined them into one dataset. We get the filtered loan-level record and saved in CSV format.

The raw dataset contains 146,809 rows and 99 columns. Each row represents a reported loan or application record, rather than a unique borrower. We save records with action_taken equal to 1, 2, or 3 respectively, leaving 104,779 eligible applications. We select candidate predictors describing applicants' financial circumstances, loan characteristics, and property features for a binary denial target. The original action_taken field is used to construct the target and is excluded from the model inputs.

Denied is the target variable, which equals 1 for denied applications (action_taken = 3) and 0 for originated loans or applications approved but not accepted (action_taken = 1 or 2). Withdrawn applications, incompleted files, purchased loans, and preapproval outcomes are excluded. The eligible dataset contains 94,870 not denied applications (90.54%) and 9,909 denied applications (9.46%). Thus, our task is a binary classification.

3 Exploratory Data Analysis

We explored the 104,779 decided applications for home purchase loans in Rhode Island from 2018 to 2024. We first removed fields that are only filled in after the lender decides (interest rate, rate spread, loan costs, purchaser type, and denial reasons), since a model would not have them for a new application. Every plot that relates a feature to the target uses only the training set (65,978 applications, a random 80% of 2018 to 2022). We looked at each column alone, pairs of features, each feature against the target, missing values, counties, and changes across years, with log axes for the skewed dollar amounts. Three findings shape our modeling plan.

First, the ratio of debt to income (DTI) has a strong, nonlinear relationship with denial (Figure 1). Applications with a DTI between 20% and 50% are denied 5% to 7% of the time, compared with 16% at 50% to 60% and 79% above 60%. The 2023 validation year shows the same pattern. Because DTI is reported as ranges at both ends, we treat it as an ordinal feature.

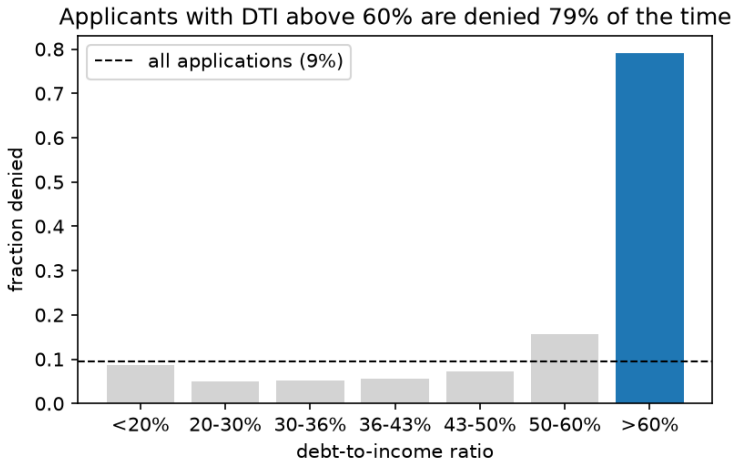


Figure 1: Denial rate for each DTI band in the training set. The dashed line is the overall rate of 9.4%.

Second, missing values carry information (Figure 2). Applications without a property value are denied 52% of the time, compared with 8.5% when it is reported, and those missing the ratio of loan to property value (LTV) or the DTI are denied 32% and 24% of the time. These are raw rates, but the property value gap also holds within every DTI band. Some applications are likely denied before an appraisal. Missing income makes little difference. We will keep indicators for missing values instead of dropping rows, and compare models with and without them, since the missingness may partly reflect the decision itself.

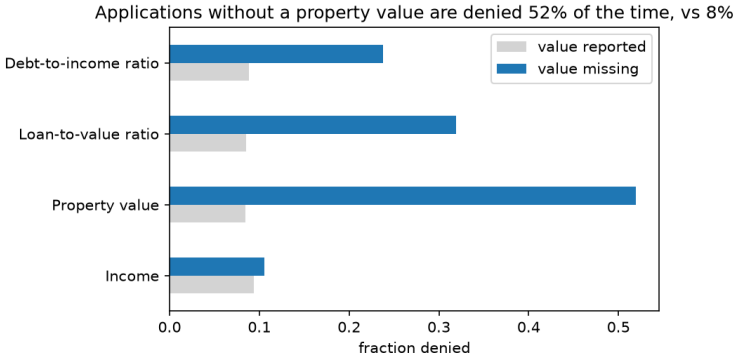


Figure 2: Denial rate when each field is reported and missing, training set.

Third, the applicants changed while the denial rate barely moved (Figure 3): 9.4% in the training years and 9.9% in 2023, a smaller change than the national rise García and Garriga [2] report. Over the same period the median loan grew from $275,000 to $375,000 by 2024, median income rose from $86,000 to $121,000, and the share of applicants with a DTI of 43% to 50% rose from 25% to 33%. We therefore keep two test sets: a random 20% of 2018 to 2022, and all of 2024 to test the model on a later year.

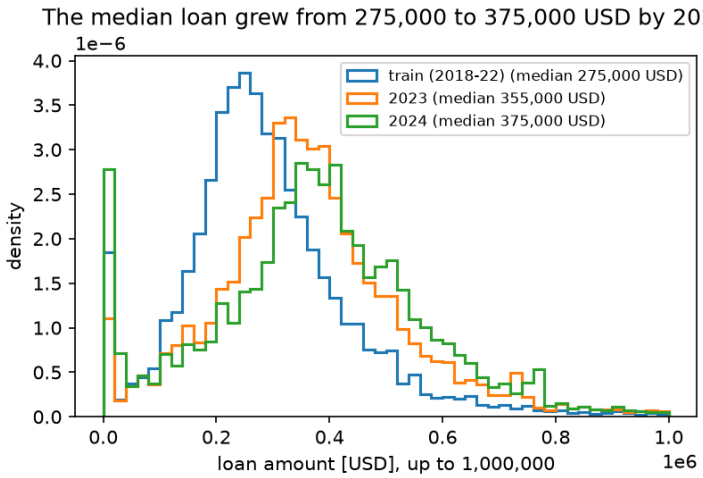


Figure 3: Loan amounts in the training years, 2023, and 2024, with each group's median in the legend.

4 Problem description and potential impact

4.1 Problem description

Our model aims to predict whether a person will get a mortgage given their information. Our target variable is binary: 0 (approved) or 1 (denied).

Several previous studies have used this dataset, many of which focus on racial equality in mortgage lending. In "Fairness by Design: Machine Learning and Interpretable Mortgage Lending", Bell et al. [4] explored a "less discriminatory alternative" for predicting mortgage approval, which maintained predictive performance while reducing disadvantages for minority groups. They shifted weight away from demographics toward DTI (debt-to-income) and LTV (loan-to-value).

In a news article "AI Exhibits Racial Bias in Mortgage Underwriting Decisions" from Lehigh University [5], they drew from a sample of 1,000 loan applications included in the 2022 HMDA dataset, to create 6,000 experimental loan applications. In the experiment, researchers manipulated race and credit score variables to determine their effects. The results show that Black applicants were more likely to be rejected or charged higher rates than otherwise identical White applicants.

Hodges et al. [6] used the yearly data from the Consumer Financial Protection Bureau under the HMDA to train several machine learning models. They sampled the data and added 10 macroeconomic indicators. They found that a deep neural network performed best, reaching an F1 score of 89%. The loan amount and the applicant income were the most important features. Unfortunately, when race and gender were left in the feature set, the model also selected them as important. Adding macroeconomic indicators did not improve the model performance.

In "Developing a novel fair-loan-predictor through a multi-sensitive debiasing pipeline: DualFair", Singh et al. [7] also trained an ML model using data from six states during 2018-2020 to predict mortgage approval. However, they achieved nearly perfect accuracy in all six states, both before and after debiasing. We could not confirm whether outcome leakage or another factor caused these

unusually high scores, so we treat their results with caution and do not use them as a performance benchmark for our model.

4.2 Potential impact

Our project will build a model that predicts whether a mortgage application in Rhode Island will be approved or denied, using the public HMDA data. We will also check whether the predictions differ greatly across racial and ethnic groups. We focus on Rhode Island because it is the community we live in. A national model usually overlooks local differences, such as local lenders and neighborhoods.

Instead of replacing the decision-making process, we aim to improve it. Potential users of our model include:

- Rhode Island lenders and credit unions - if the model predicts a denial, the mortgage application can get a second human review
- Regulators and fair-housing organizations, since HMDA is an Act about fair-housing for minorities
- Housing counselors and applicants - counselors can run the model for their clients, and if it returns a denial, they can look into how to improve the client's profile

References

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[6] H. Hodges, C. Garrity, and J. Pope, "Deep learning, feature selection and model bias with home mortgage loan classification," in *Proc. 13th Int. Conf. Pattern Recognition Applications and Methods (ICPRAM)*, vol. 1, 2024, pp. 248–255, doi: 10.5220/0012326800003654.

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A Feature and Column Descriptions

This appendix describes the candidate predictors selected in our current preprocessing code, the derived binary target, and two supporting fields. The predictor definitions refer to the original variables before encoding. These are the fields retained for the current analysis; the raw API downloads contain 99 columns.

A.1 Candidate Predictors

Table 1: Definitions, measurement units, and category codes for the ten candidate predictors before encoding.

Predictor	Type	Description
`loan_amount`	Numerical	Requested or originated loan amount, in US dollars.
`income`	Numerical	Gross annual income relied on in the credit decision or application review, in thousands of US dollars.
`loan_to_value_ratio`	Numerical	Total property-secured debt relative to property value, expressed as a percentage.
`property_value`	Numerical	Property value used in the credit decision, in US dollars.
`loan_term`	Numerical	Loan duration in months; for example, 360 months represents 30 years.
`debt_to_income_ratio`	Ordered categorical	Monthly debt relative to monthly income, expressed as a percentage. The ordered levels are defined in Table 2.
`loan_type`	Categorical	1 = conventional; 2 = Federal Housing Administration (FHA) insured; 3 = Veterans Affairs (VA) guaranteed; 4 = USDA Rural Housing Service (RHS) or Farm Service Agency (FSA) guaranteed.
`occupancy_type`	Categorical	1 = principal residence; 2 = second residence; 3 = investment property.
`construction_method`	Categorical	1 = site-built dwelling; 2 = manufactured home.
`total_units`	Categorical	Number of dwelling units: 1, 2, 3, 4, 5–24, 25–49, 50–99, 100–149, or > 149. Values above four are reported as grouped ranges.

A.2 Debt-to-Income Category Order

The valid debt-to-income levels are ordered from the lowest to the highest ratio. The order numbers below document the ordering and do not indicate that this encoding has already been applied to the exported CSV files. Percent signs describe the measurement unit; individual percentage levels may appear as numeric strings in the CSV.

Table 2: Valid debt-to-income levels in increasing order.

Order	Category	Meaning
1	$< 20\%$	Below 20 percent
2	20%–$< 30\%$	At least 20 percent and below 30 percent

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Table 2 (continued)

Order	Category	Meaning
3	30%–< 36%	At least 30 percent and below 36 percent
4	36%	36 percent
5	37%	37 percent
6	38%	38 percent
7	39%	39 percent
8	40%	40 percent
9	41%	41 percent
10	42%	42 percent
11	43%	43 percent
12	44%	44 percent
13	45%	45 percent
14	46%	46 percent
15	47%	47 percent
16	48%	48 percent
17	49%	49 percent
18	50%–60%	50 through 60 percent, inclusive
19	> 60%	Above 60 percent

Blank entries, `NA`, and `Exempt` are unavailable states and have no position in this ordering.

A.3 Target and Supporting Fields

Table 3: The derived target and the two supporting fields retained during preparation.

Column	Role	Description
`denied`	Binary target	1 = denied; 0 = not denied. Constructed from `action_taken` according to Table 4.
`action_taken`	Source outcome	Original outcome code. Used for eligibility and target construction, then excluded from predictors to prevent target leakage.
`activity_year`	Splitting field	Calendar reporting year: 2018, 2019, 2020, 2021, 2022, 2023, or 2024. Stored as text and used to select the periods for splitting; excluded from predictors.

Table 4: Treatment of the original `action_taken` codes in this study.

Code	Outcome	Treatment
1	Loan originated	Retain; `denied` = 0
2	Application approved but not accepted	Retain; `denied` = 0
3	Application denied	Retain; `denied` = 1
4	Application withdrawn by applicant	Exclude

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Table 4 (continued)

Code	Outcome	Treatment
5	File closed for incompleteness	Exclude
6	Purchased loan	Exclude
7	Preapproval request denied	Exclude
8	Preapproval request approved but not accepted	Exclude

The processing DataFrame contains 13 columns: ten candidate predictors, `denied`, `action_taken`, and `activity_year`. The exported modeling files contain 11 columns: the ten candidate predictors and `denied`. The two supporting fields are excluded from the model inputs.

A.4 Unavailable Values

The current code flags actual missing values and the strings `""`, `NA`, and `Exempt` as unavailable. For numerical predictors, these string markers are replaced with `NaN` before numeric conversion. Categorical predictors retain their string labels at this stage. Imputation, missing-value indicators, and encoding are separate preprocessing decisions.

A.5 Source Documentation

Field definitions and category meanings follow the official HMDA documentation: `https://ffiec.cfpb.gov/documentation/publications/loan-level-datasets/lar-data-fields`.
The complete raw schema is available at `https://ffiec.cfpb.gov/documentation/publications/loan-level-datasets/public-lar-schema`.
This appendix uses the names in the downloaded CSV. In particular, `loan_to_value_ratio` corresponds to `combined_loan_to_value_ratio` in the official documentation.

B Additional Exploratory Analysis

Here we add two results that support Section 3: a scatter matrix of the numerical features colored by outcome, and a summary of the applications by year. Any result that uses the outcome is based on the training set only, and we did not use the 2024 labels.

B.1 Scatter Matrix by Outcome

Figure 4 shows the five numerical features for a random sample of 10,000 training applications with no missing values. Loan amount and property value lie close to a line, since LTV is roughly one divided by the other. Denied applications cluster at an LTV of about 95% to 100% and at high DTI. DTI shows up as a few vertical bands because we replaced each range at the two ends with a single value inside it.

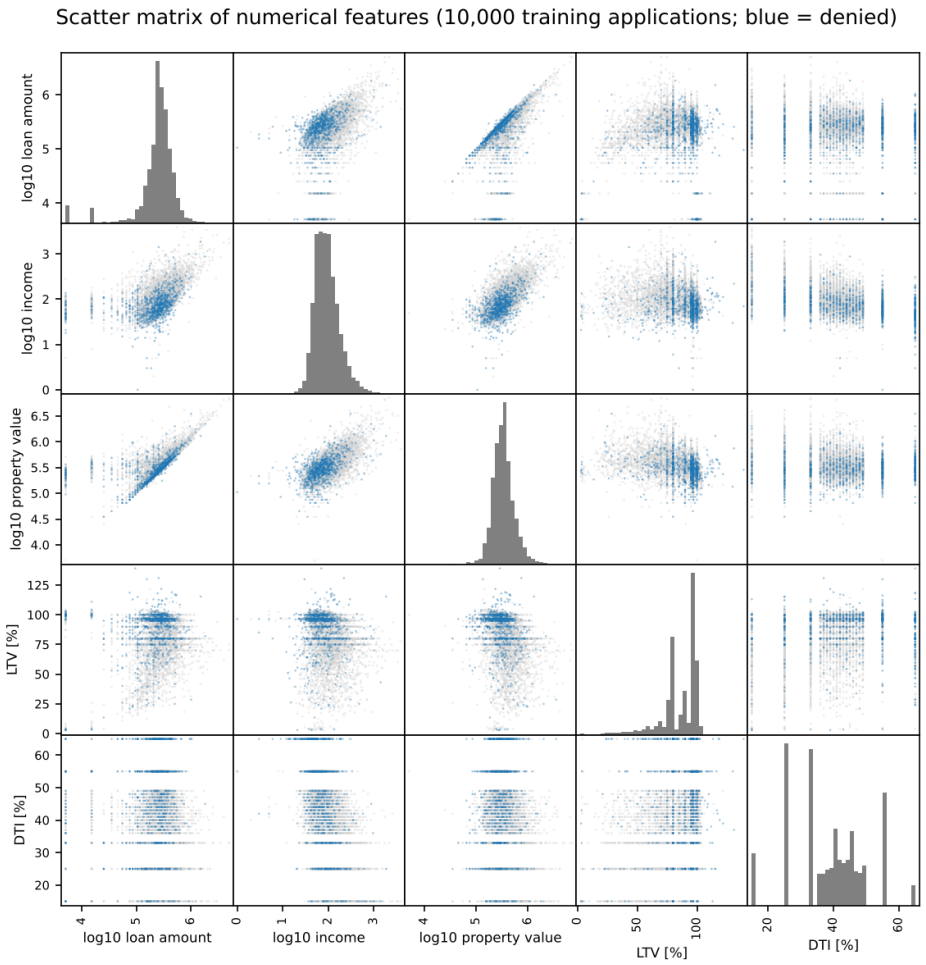


Figure 4: Scatter matrix of the numerical features for 10,000 training applications. Loan amount, income, and property value are on a log scale. Blue points are denied applications and gray points are not denied.

B.2 Summary by Year

Median loan amounts and incomes went up every year from 2018 to 2024 (Table 5). The FHA share dropped from about 24% to 18%, and the share of applicants with a DTI of 43% to 50% rose from about 25% before 2022 to 33% in 2024. Over the same period, the denial rate stayed between 8% and 11%.

Table 5: Eligible applications by year (median income in thousands of US dollars).

Year	Applications	Median loan	Median income	FHA	DTI 43 to 50%	Manufactured	Denied
2018	15,965	225,000	79	24.1%	26.0%	0.9%	11.0%
2019	16,234	245,000	82	24.1%	24.2%	0.9%	9.9%
2020	17,003	265,000	84	24.9%	24.6%	0.6%	8.5%
2021	18,402	305,000	87	24.1%	24.6%	0.7%	8.2%

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Table 5 (continued)

Year	Applications	Median loan	Median income	FHA	DTI 43 to 50%	Manufactured	Denied
2022	14,869	335,000	98	21.9%	27.5%	0.7%	9.8%
2023	10,630	355,000	114	22.1%	30.6%	1.2%	9.9%
2024	11,676	375,000	121	18.4%	33.0%	1.3%	n/a